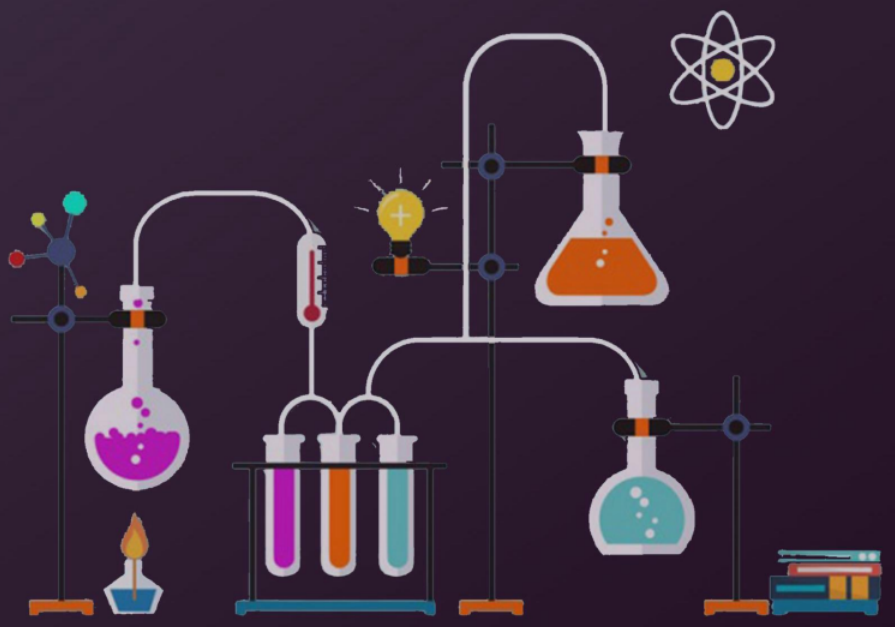
CHEMISTRY

Haloalkane and Haloarenes



SHEET

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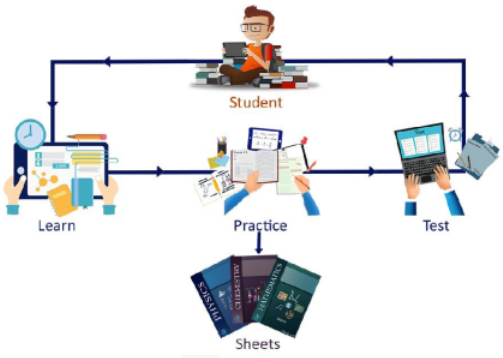
Contents

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Introduction to Haloalkenes and Haloarenes				
2	Basics of Reaction Mechanism				
3	Electrophile, Nucleophile and Role of Solvent	1-2			
4	Nucleophilicity, Basic Strength and Leaving Groups	3-4	1	1-2	1
5	Preparation Methods Recap	5-6		3-5	2, 12
6	Nucleophilic Substitution Reaction Via SN1	6-9	2	6	13
7	Nucleophilic Substitution Reaction Via SN2	10-12	3-6, 12-13, 16-18	7-16	3
8	Preparation of Alkyl Halides by Nucleophilic Substitution	13-14	7-8	17-18	10
9	Elimination Reactions	15-17		19-25	9, 11
10	SN1 vs SN2, E1 vs E2, Sustitution vs Elimination		9		
11	Chemical and Physical properties of Haloarenes				
12	ACC	18-21	14-15, 19	26-30	4-8, 14-15

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Electrophile, Nucleophile and Role of Solvent

Q1. $\overset{\oplus}{C}H_3$, $\overset{\ominus}{C}H_3$ and $\dot{C}H_3$ respectively are :-

(A) Electrophile, nucleophile, nucleophile
(B) Electrophile, nucleophile, electrophile
(C) Nucleophile, nucleophile, nucleophile
(D) Electrophile, electrophile, electrophile

Q2. Which one of the following is an electrophile:-

(A) RO^- (B) BF_3
(C) NH_3 (D) ROH

Nucleophilicity, Basic Strength and Leaving Groups

Q3. Order of nucleophilicity in polar aprotic solvent:-

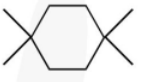
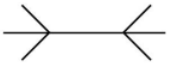
(A) $I^- > Br^- > Cl^- > F^-$
(B) $F^- > Cl^- > Br^- > I^-$
(C) $I^- > Cl^- > Br^- > F^-$
(D) $F^- > I^- > Br^- > Cl^-$

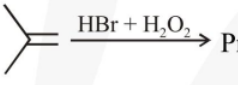
Q4. The order of ease of leaving group would be :-

–OAc	–OMe
I	II
$-OSO_2Me$	$-OSO_2CF_3$
III	IV

(A) I > II > III > IV
(B) IV > III > I > II
(C) III > II > I > IV
(D) II > III > IV > I

Q5. Choose that alkane which cannot give only one monochloro derivative upon reaction with chlorine in sun light :

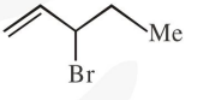
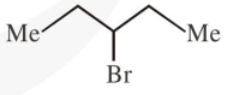
(A)  (B) 
(C)  (D) 

Q6.  Product. , Product is :

(A) 3° alkyl bromide (B) 1° alkyl bromide
(C) 2° alkyl bromide (D) 1° alcohol

Nucleophilic Substitution Reaction Via SN1

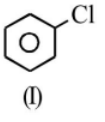
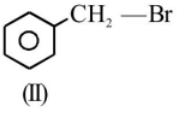
Q7. Consider the following bromides :

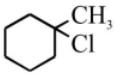
(I) 
(II) 
(III) 

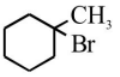
The correct order of S_N1 reactivity is :

(A) I > II > III (B) II > III > I
(C) II > I > III (D) III > II > I

Q8. Reactivity order towards SN^1 is


(I)

(II)

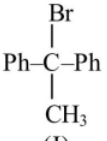

(III)

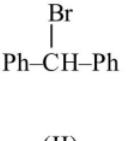

(IV)

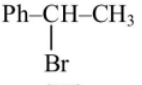
(A) II > I > IV > III (B) IV > III > I > II

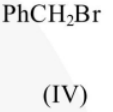
(C) VI > III > I > II (D) II > IV > III > I

Q9. The decreasing order of rate of S_N1 reaction for the following compounds is -


(I)


(II)


(III)


(IV)

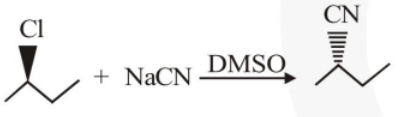
(A) (I) > (II) > (III) > (IV)

(B) (II) > (I) > (III) > (IV)

(C) (IV) > (III) > (II) > (I)

(D) (III) > (IV) > (II) > (I)

Nucleophilic Substitution Reaction Via SN2

Q10. 

Identify mechanism of above reaction

(A) SN^2 (B) SN^1

(C) SN^i (D) E^2

Q11. 1°gem, 2°gem and vic-dihalide of propane on treatment with aqueous KOH respectively gives:-

(A) Propanol, propanone, propanoic acid

(B) Propanone, propanol, glycol

(C) Propanal, propanone, propylene glycol

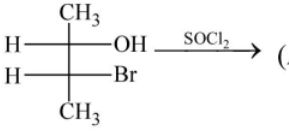
(D) Propanal, propanone, trimethylene glycol

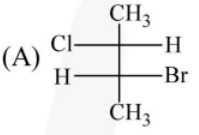
Q12. $CH \equiv CH \xrightarrow[1\ eq.]{NaNH_2} A \xrightarrow{CH_3-Cl} B$ is :

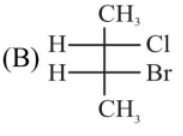
(A) $CH_3-C\equiv C-CH_3$ (B) $CH_3C\equiv CH$

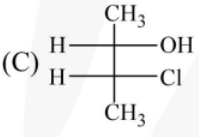
(C) $HC\equiv CH$ (D) $HC\equiv C-C\equiv CH$

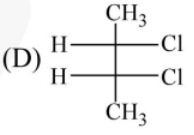
Preparation of Alkyl Halides by Nucleophilic Substitution

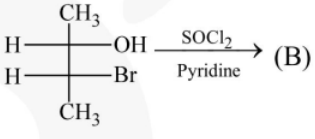
Q13. 

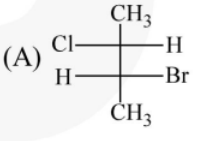
(A) 

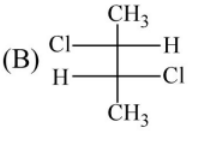
(B) 

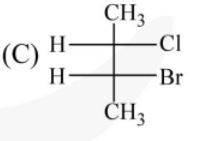
(C) 

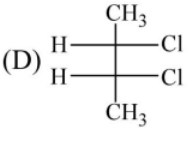
(D) 

Q14. 

(A) 

(B) 

(C) 

(D) 

Elimination Reactions

Q15. Alcoholic caustic potash is a specific reagent for :-

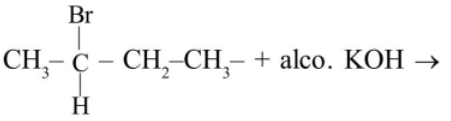
(A) Dehydration

(B) Dehydrogenation

(C) Decarboxylation

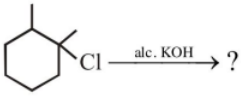
(D) Dehydrohalogenation

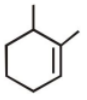
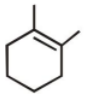
Q16. Major product of the following reaction is:-

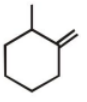


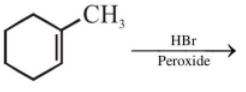
(A) 1-Butene (B) 2-Butene
(C) Butane (D) 1-Butyne

Q17. Possible products of the reaction are :-

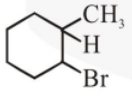
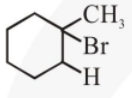


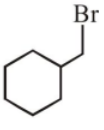
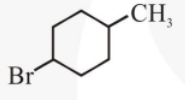
(A)  (B) 

(C)  (D) All of these

Q18. 

Major product will be:

(A)  (B) 

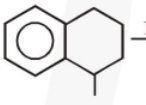
(C)  (D) 

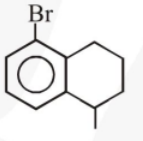
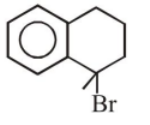
Q19. N-bromosuccinimide is the useful reagent and when reacts with $CH_3–CH = CH_2$:-

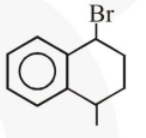
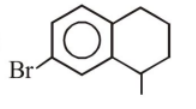
(A) The allylic hydrogen is replaced by Br atom
(B) Br atom adds on double bond
(C) The hydrogen atom present on unsaturated C-atom is repalced by Br-atom
(D) None of the above

Q20. Iodoethane reacts with sodium in the presence of dry ether :-

(A) Pentane (B) Propane
(C) Butene (D) Butane

Q21.  X (major product) ; X is:

(A)  (B) 

(C)  (D) 

Exercise - 1A *Concept Builder*

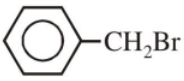
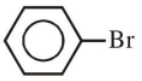
ONE OR MORE THAN ONE CORRECT TYPE

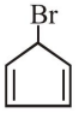
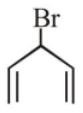
*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. The nucleophilicities of CH_3^-, NH_2^-, OH^- and F^- decrease in the order -

(A) $CH_3^- > NH_2^- > OH^- > F^-$
(B) $OH^- > NH_2^- > CH_3^- > F^-$
(C) $NH_2^- > OH^- > CH_3^- > F^-$
(D) $CH_3^- > OH^- > F^- > NH_2^-$

Q2. Which of the following compound can't give S_N1 reaction readily ?

(A)  (B) 

(C)  (D) 

Q3. Which of the following statements is/are **true**?

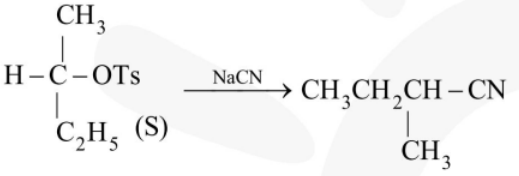
(A) $CH_3–CH_2–CH_2–I$ will react more readily than $(CH_3)_2$ CHI for S_N2 reactions.

(B) $CH_3–CH_2–CH_2–Cl$ will react more readily than $CH_3–CH_2–CH_2–Br$ for S_N2 reaction.

(C) $CH_3–CH_2–CH_2–CH_2–Br$ will react more readily than $(CH_3)_3C–CH_2–Br$ for S_N2 reactions

(D) $CH_3–O–C_6H_5–CH_2$ Br will react more readily than $NO_2–C_6H_5–CH_2Br$ for S_N2 reaction

Q4. Consider the given reaction



which of following statements are correct for above reaction.

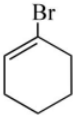
(A) Product formation takes place due to the breaking of O–Ts

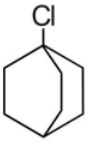
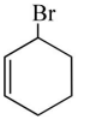
(B) The reaction is S_{N^2}

(C) The reaction is S_{N^1}

(D) Configuration of product is (R)

Q5. Rate of S_N2 will be negligible in

(A)  (B) 

(C)  (D) 

Q6. Aryl halides are less reactive towards nucleophilic substitution reaction as compared to alkyl halides due to -

(A) The formation of less stable carbonium ion

(B) Resonance stabilization

(C) The inductive effect

(D) sp^2 hybridised carbon attached to the halogen

Q7. Which reactions giving alkyl chloride from alcohol

(A) $EtOH + SOCl_2 \longrightarrow$

(B) $EtOH + SOCl_2 + Pyridine \longrightarrow$

(C) $EtOH + PCl_3 \longrightarrow$

(D) $Me_3COH + HCl + ZnCl_2 \longrightarrow$

Q8. A gem dichloride is formed in the reaction :

(A) CH_3CHO and PCl_5

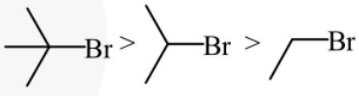
(B) CH_3COCH_3 and PCl_5

(C) $CH_2 = CH_2$ and Cl_2

(D) $CH_2 = CHCl$ and HCl

Q9. Correct statement is -

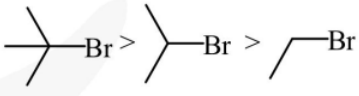
(A) Order of reactivity in E_1 mechanism



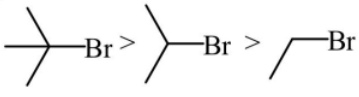
(B) Order of nucleophilicity

$F^- < Cl^- < Br^- < I^-$ (in H_2O)

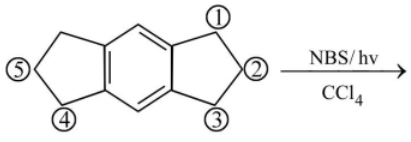
(C) Order of reactivity in S_N2 mechanism



(D) Order of reactivity in S_N1 mechanism



Q10. Bromination can take place at



(A) 1 (B) 2 (C) 3 (D) 4

Q11. Toluene, when treated with Br_2/ Fe, gives p-bromotoluene as the major product, because the CH_3 group -

(A) is para directing

(B) is meta directing

(C) activates the ring by hyperconjugation

(D) deactivates the ring

PARAGRAPH TYPE

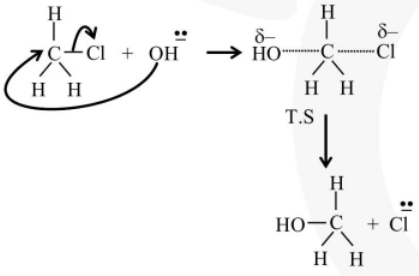
*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONE OR MORE THAN ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q. 12 to 13

Consider the following reaction -

$$CH3–Cl + \ddot{O}H \longrightarrow CH3–OH + \ddot{Cl}$$

Experimentally it has been found that rate of this reaction depends on the concentration of CH3–Cl and replaces Cl^- i.e. one nucleophile replaces another so this reaction is known as nucleophilic substitution bimolecular reaction.



Q12. Choose the correct statement for the reaction given in passage

(A) solvent like DMF and DMSO favour this reaction

(B) It is a Bimolecular reaction

(C) On doubling the concentration of CH3–Cl rate becomes four times

(D) Both A & B

Q13. Rate of S_N2 depends on

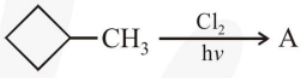
(A) Conc of Nucleophile

(B) Conc of substrate

(C) Nature of leaving group

(D) Nature of solvent

Paragraph for Q. 14 to 15



Q14. Which of following is/are correct :

(A) All products are optically active

(B) All products are optically inactive

(C) It gives two enantiomer

(D) It give 2 pair of geometrical isomerism

Q15. Number of products which is separate by fractional distillation is :

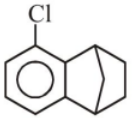
(A) 2 (B) 4

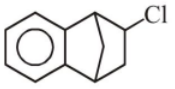
(C) 1 (D) 3

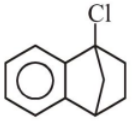
MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q16.

List-I	**List-II**
(P) 	(1) S_N1 reaction may takes place

(Q)  (2) S_N2 reaction may takes place

(R)  (3) S_N1 is not possible

(S)  (4) S_N2 is not possible

Code :

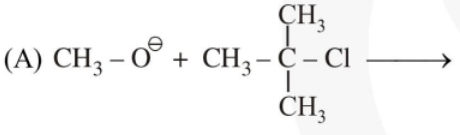
	P	Q	R	S
(A)	3, 4	1, 2	3, 4	3, 4
(B)	3, 4	2, 2	1, 2	4, 4
(C)	4, 4	2, 2	1, 1	3, 3
(D)	4, 4	1, 1	2, 2	3, 3

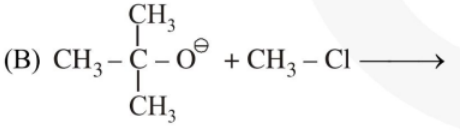
MATCH THE COLUMN TYPE

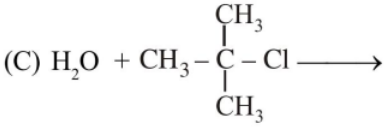
*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

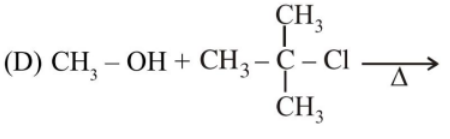
Q17. Match the four reactions with type of mechnism given in List II with the corresponding reaction provided in List - I and select the correct answer given below in lists.

Column - I **Column - II**

(A)  (P) SN^2

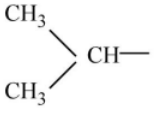
(B)  (Q) E^2

(C)  (R) SN^1

(D)  (S) E^1

Q18. Match Column I with Column II for given S_N2 reaction & select the correct answer from the codes given below :

$$Z\text{–}CH_2Br + CH_3O^{\ominus} \longrightarrow Z\text{–}CH_2\text{–}OCH_3 + Br^{\ominus}$$

Column I (Z)	Column II (Rate of SN_2)
(A) H–	(P) 0.1
(B) CH_3–	(Q) 3
(C) C_2H_5–	(R) 1
(D) 	(S) 100

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q19. How many total products possible when 2-chloro butane reacts with alcoholic potash.

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The leaving group ability of the following groups in a nucleophilic acyl substitution by a given nucleophile decreases in the order:-

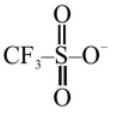
(A) $–NH_2 > –Cl > –OCOR > –OR$

(B) $–Cl > –OCOR > –OR > –NH_2$

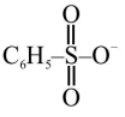
(C) $–OCOR > –OR > –Cl > –NH_2$

(D) $–OR > –Cl > – NH_2 > –OCOR$

Q2. Consider the following nuecleophiles :



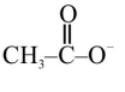
(I)



(II)



(III)



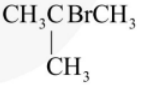
(IV)

When attached to sp^3 hybridised carbon, their leaving group ability in nucleophilic substitution reactions decreases in the order.

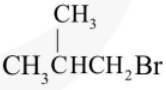
(A) I > II > III > IV (B) I > II > IV > III

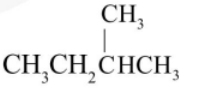
(C) IV > I > II > III (D) IV > III > II > I

Q3. The reaction of HBr with  in the presence of peroxide will give :-

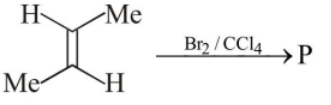
(A) 

(B) $CH_3CH_2CH_2CH_2Br$

(C) 

(D) 

Q4. Select incorrect statements about the product (P) of the reaction :



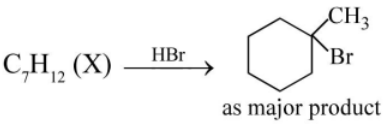
(A) P is optically inactive due to internal compensation

(B) P is optically inactive due to the presence of plane of symmetry in the molecule

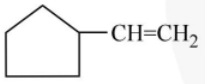
(C) The structure of P can have three optical isomers possible.

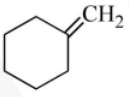
(D) P can have four possible optical isomers.

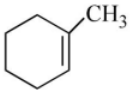
Q5. In the given reaction

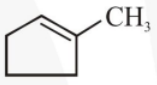


(X) can not be :

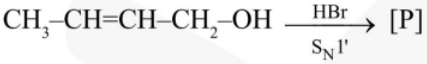
(A) 

(B) 

(C) 

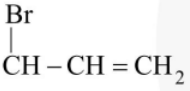
(D) 

Q6. Consider the given reaction:

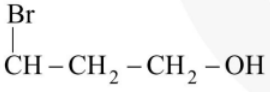


In the given reaction the product [P] is :

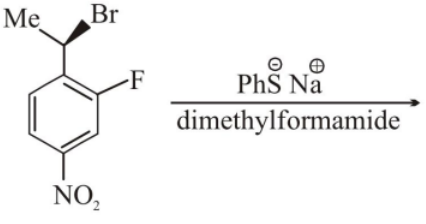
(A) $CH_3–CH=CH–CH_2–Br$

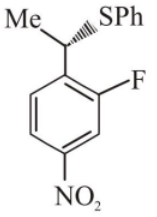
(B) 

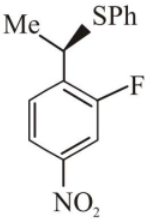
(C) $CH_2=CH–CH=CH_2$

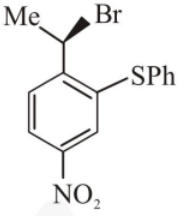
(D) 

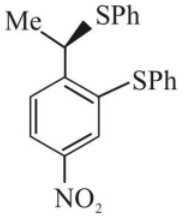
Q7. The major product of the following reaction is -



(A) 

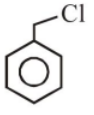
(B) 

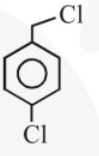
(C) 

(D) 

Q8. Which of the following is most reactive toward SN^2 reaction ?

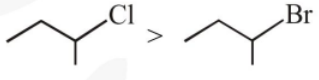
(A) 

(B) 

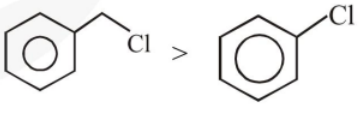
(3) 

(D) 

Q9. Which of the following represent correct reactivity order for SN^1 as well as SN^2 mechanism -

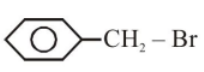
(A) 

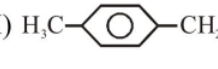
(B) $CH_2 = CHCl > CH_3CH_2–Cl$

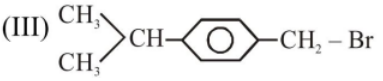
(C) 

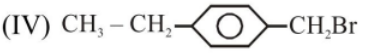
(D) $CH_2 = CHCl > CH_2 = CH–CH_2Cl$

Q10. Arrange the following compounds in order of decreasing rate of hydrolysis for SN^1 reaction:-

(I) 

(II) 

(III) 

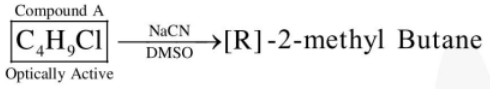
(IV) 

(A) II > IV > III > I

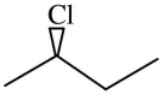
(B) III > IV > II > I

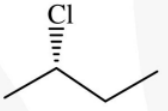
(C) IV > III > II > I

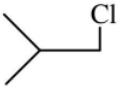
(D) III > II > IV > I

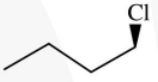
Q11. 

Nitrile; Identify compound 'A' ?

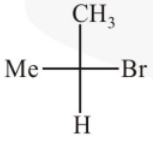
(A) 

(B) 

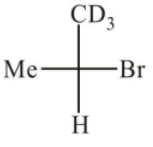
(C) 

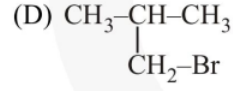
(D) 

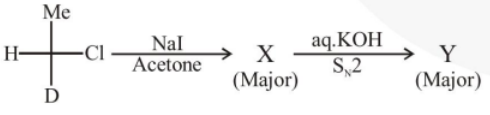
Q12. Which of the following compounds will give racemic mixture by S_N1 reaction ?

(A) 

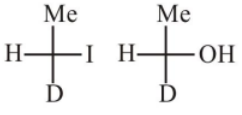
(B)

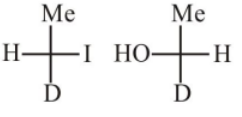
(C) 

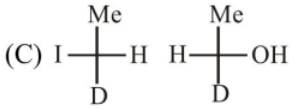
(D) 

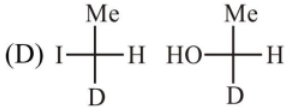
Q13. 

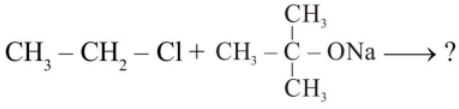
X & Y should be :

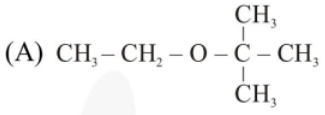
(A) 

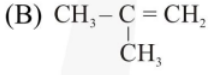
(B) 

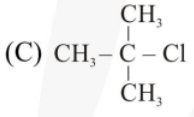
(C) 

(D) 

Q14. 

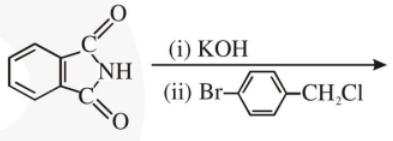
(A) 

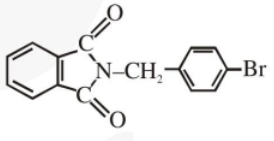
(B) 

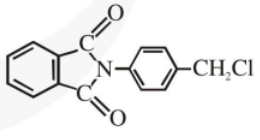
(C) 

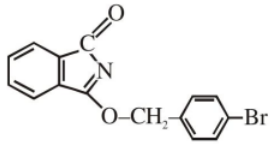
(D) $CH_3 - CH_2 - OH$

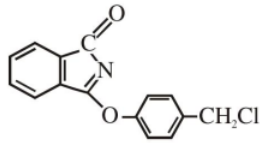
Q15. The major product of the following reaction is -



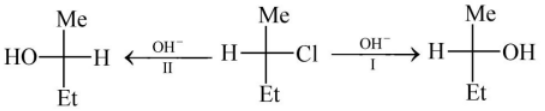
(A) 

(B) 

(C) 

(D) 

Q16.



B A

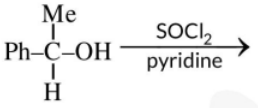
Mechanism involved :

(A) I can't be S_N1

(B) II can't be S_N2

(C) I S_N1 & II S_N2

(D) I S_N2 & II S_N1

Q17. 

Which statement is true for the above reaction -

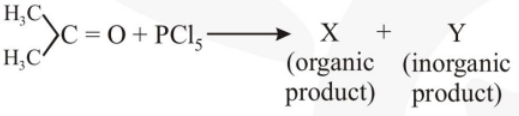
(A) Retention of configuration

(B) Inversion of configuration

(C) Retention and inversion both

(D) None of these

Q18. Which of the following option is correct regarding X & Y.



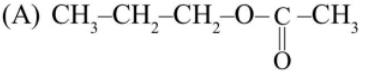
(A) X is vicinal dihalide

(B) X is gemdihalide

(C) Y having central atom which is sp^3d hybridised

(D) In Y, phosphorus has +3 oxidation state

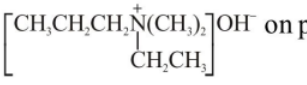
Q19. Which ester on pyrolysis gives isobutylene:-

(A) 

(B) $CH_3COOCH_2CH_2CH_2CH_3$

(C) $CH_3COOCH_2–CH(CH_3)_2$

(D) All the above

Q20.  on pyrolysis gives :-

(A) Propene (major) + ethene

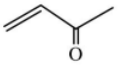
(B) Ethene (major) + propene

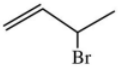
(C) Ethene : propene :: 1 : 1 (product)

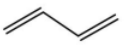
(D) None

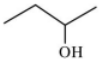
Q21. Identify (C) product of following reaction ?



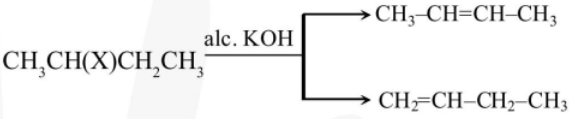
(A) 

(B) 

(C) 

(D) 

Q22. For the reaction

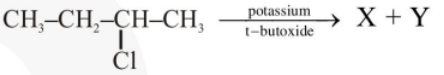


(A) $CH_3–CH=CH–CH_3$ predominates.

(B) $CH_2=CH–CH_2–CH_3$ predominates.

(C) Both are formed in equal amounts.

(D) The product ratio is dependent on the halogen X.

Q23. 

X and Y in the above set are expected as :-

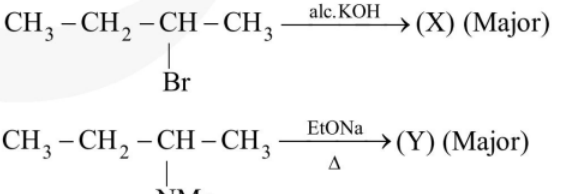
(A) 2-Butanol and 1-butene

(B) 2-Butene and 1-butanol

(C) 2-Butene and 1-butene

(D) 1-Butanol and 1-butene

Q24.



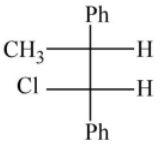
Product (X) & (Y) respectively is -

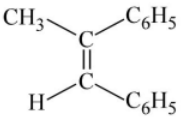
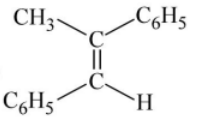
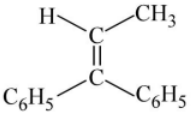
(A) 1-butene, trans-2-butene

(B) 1-butene, cis-2-butene

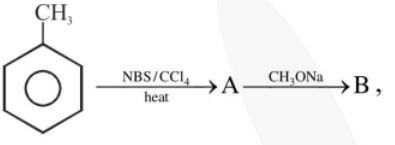
(C) cis-2-butene, 1-butene

(D) trans-2-butene, 1-butene

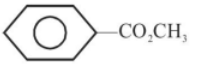
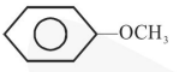
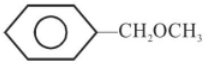
Q25.  $\xrightarrow{\text{alc.KOH}}$ A, A is -

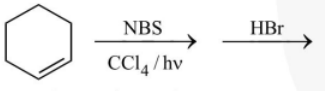
(A) 
(B) 
(C) 
(D) None is correct

Q26. The final product (B) formed in the reaction sequence :-

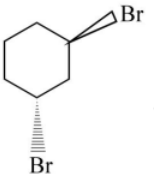
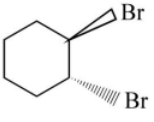
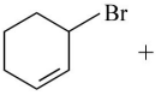


product (B) is -

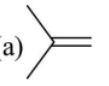
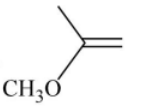
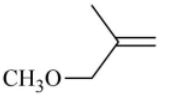
(A) 
(B) 
(C) 
(D) 

Q27. 

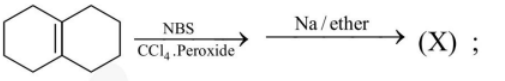
enantiomeric pair

(A)  + enantiomer
(B)  + enantiomer
(C)  + enantiomer
(D) None of these

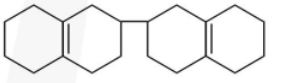
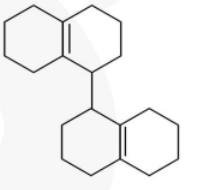
Q28. What is the order of reaction with HBr ?

(a) 
(b) 
(c) 

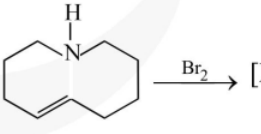
(A) a > b > c
(B) b > a > c
(C) c > b > a
(D) b > c > a

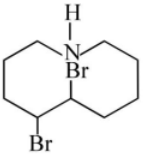
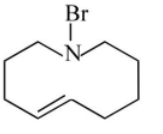
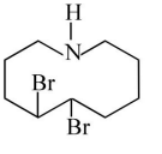
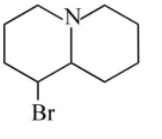
Q29. 

X is -

(A) 
(B) 
(C) 
(D) None of these

Q30. In the given reaction :

 [X], [X] is :

(A) 
(B) 
(C) 
(D) 

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ONE OR MORE THAN ONE of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.

Q1. Which of the following option is/are **not correct** regarding $CH_3S^{\ominus}$ and $CH_3O^{\ominus}$:

(A) $CH_3O^{\ominus}$ is stronger base than $CH_3–S^{\ominus}$

(B) $CH_3O^{\ominus}$ is stronger nucleophile than $CH_3S^{\ominus}$ (in H_2O)

(C) $CH_3O^{\ominus}$ is weaker base than $CH_3S^{\ominus}$.

(D) $CH_3O^{\ominus}$ & $CH_3S^{\ominus}$ both give elimination product when react with $CH_3–\underset{\displaystyle CH_3}{\underset{|}{CH}}–Cl$

Q2. Products formed when HCl adds to 2,4-hexadiene is :

(A) 4-chloro-2-hexene

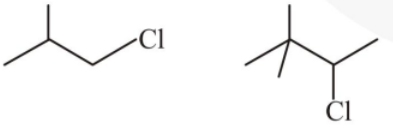
(B) 2-chloro-3-hexene

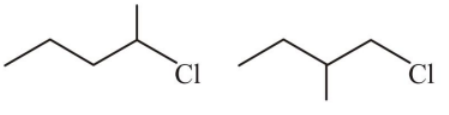
(C) 2-chloro-4-hexene

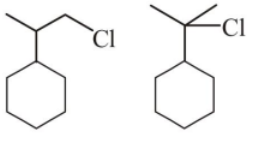
(D) 1-chloro-2-hexene

Q3. In the given pair in which pair the first compound is more reactive than second towards S_N2 reaction.

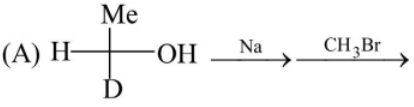
(A) 

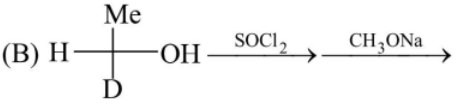
(B) 

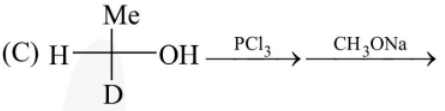
(C) 

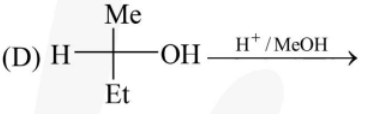
(D) 

Q4. In which of the following reaction configuration about chiral C is retained in the final product :

(A) 

(B) 

(C) 

(D) 

PARAGRAPH TYPE

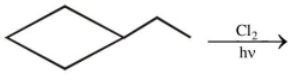
*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONE OR MORE THAN ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q. 5 to 6

Radicals are obtained by homolysis of bond. Since energy is required to break a bond in homolytic pattern, hence light is a possible energy source and ultravoilet light with an associated energy of 586 KJ/mole ($h\nu$) is used for homolysis. It can decompose many organic compounds including DNA in skin cells.

During halogenation in presence of ultravoilet light a free radical substitution reaction takes place.

A reaction is given as–



Q5. Total number of possible monochloride obtained.

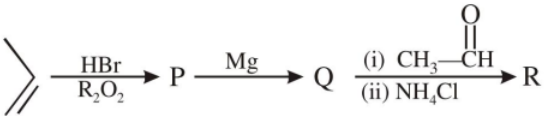
(A) 5 (B) 9 (C) 10 (D) 6

Q6. Total number of fractions obtained on fractional distillation of product mixture :

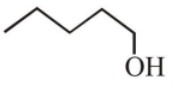
(A) 7 (B) 5 (C) 8 (D) 6

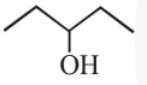
Paragraph for Q. 7 to 8

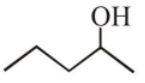
In given reaction sequence :

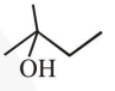


Q7. Product 'R' is :

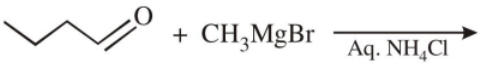
(A) 

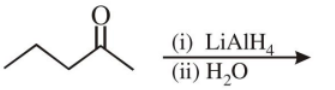
(B) 

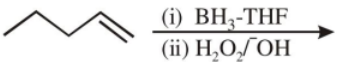
(C) 

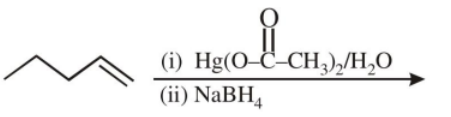
(D) 

Q8. Product R can't obtained as a major product in reaction :

(A) 

(B) 

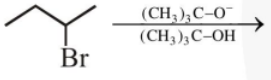
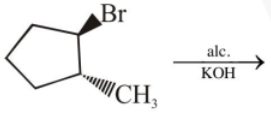
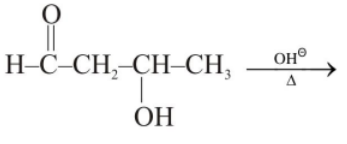
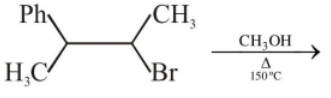
(C) 

(D) 

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q9.

List-I (Reaction)	**List-II (Major product / mechanism preferred)**
(P) 	(1) $E_{1\,CB}$ Mech.
(Q) 	(2) E_2 Mech.
(R) 	(3) E_1 Mech.
(S) 	(4) Saytzeff product

Code :

	P	Q	R	S
(A)	3	2	4	1
(B)	2	2	1	3
(C)	2	3	1	4
(D)	1	4	3	2

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q10.

Column - I	**Column - II**
(A) $CH_3Br + NaI \xrightarrow[\text{acetone}]{\text{dry}}$	(P) Markownikov product is formed
(B) $R—Br + AgF \longrightarrow$	(Q) Finkelstein Reaction
(C) $CH_3CH = CH_2 + HCl \longrightarrow$	(R) Swarts Reaction
(D) $CH_3–CH = CH_2 + HOCl \longrightarrow$	(S) Substitution Reaction

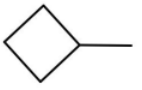
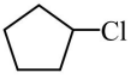
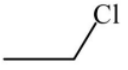
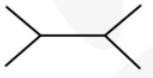
Q11. Match the Column I with Column II and select the correct answer using the codes given below the Lists.

Column I	**Column II**
(A) $CF_3–CHCl_2 \xrightarrow{\text{alc.KOH}/\Delta} CF_2 = CCl_2$	(P) Elimination Reaction
(B) $CH_3-\overset{\overset{CH_3}{\vert}}{\underset{\underset{CH_3}{\vert}}{C}}-OH \xrightarrow{H^{\oplus}} CH_3-\underset{\underset{CH_3}{\vert}}{C}=CH_2$	(Q) Carbocation
(C) $CH_3–CH_2–Br \xrightarrow[\Delta]{\text{alc.KOH}} CH_2 = CH_2$	(R) Carbanion
(D) $CH_3-\overset{\overset{Br}{\vert}}{\underset{\underset{CH_3}{\vert}}{C}}-CH_3 \xrightarrow[\Delta]{EtOH} CH_3-\underset{\underset{CH_3}{\vert}}{C}=CH_2$	(S) Free radical

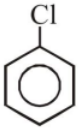
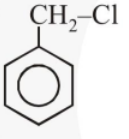
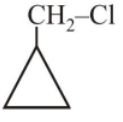
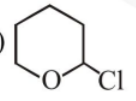
NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q12. Number of compound which form single monochloroderivatives on monochlorination.

(i) 
(ii) 
(iii) 
(iv)
(v) 
(vi) 
(vii) 
(viii) 
(ix) 
(x) 

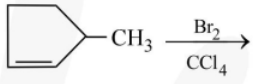
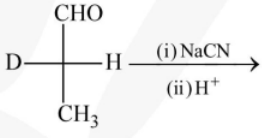
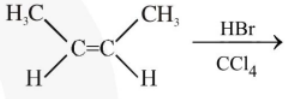
Q13. How many of following in compare with 2-chloro propane give fast SN1 reaction:

(A) 
(B) 
(C) 
(D) $CH_2=CH-Cl$
(E) 
(F) 

SUBJECTIVE TYPE

Q14. A primary alkyl bromide (A), C_4H_9Br, reacted with alcoholic KOH to give compound (B). Compound (B) reacted with HBr to give an isomer of (A). When (A) was reacted with sodium metal it gave compound (D), C_8H_{18}, which was different from the compound produced when n-butyl bromide was reacted with sodium. Draw the structure of (A) and write equations for all the reactions.

Q15. Which of following reaction products are diastereomer of each other :

(A) 
(B) 
(C) 
(D) $CH_3-\underset{\underset{Et}{|}}{CH}-CH=CH-Ph \xrightarrow[peroxide]{HCl}$
(Optically pure)

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
B	B	B	B	C	B	B	D	A	A
11	12	13	14	15	16	17	18	19	20
C	B	B	A	D	B	D	A	A	D
21									
B									

Ex-1A

1. A **2.** B,C **3.** A,C **4.** B,D **5.** A,B,C

6. B,D **7.** A,B,C,D **8.** A,B,D **9.** A,B,D **10.** A,C,D

11. A,C **12.** D **13.** A,B,C,D **14.** C,D **15.** A

16. A **17.** (A) → Q; (B) → P ; (C) → R ; (D) → S

18. (A)→S ; (B)→Q ; (C)→R ; (D)→P **19.** 3

Ex-2									
1	2	3	4	5	6	7	8	9	10
B	B	C	D	D	B	A	D	C	A
11	12	13	14	15	16	17	18	19	20
A	C	C	A	A	C	B	B	C	B
21	22	23	24	25	26	27	28	29	30
C	A	C	D	A	D	A	B	C	D

Ex-2A

1. B,C.D **2.** A,B **3.** B,D **4.** A,C **5.** C

6. A **7.** C **8.** C **9.** B

10. (A) → Q, S ; (B) → R, S ; (C) → P ; (D) → P

11. (A) → P, R ; (B) → P, Q ; (C) → P ; (D) → P, Q

12. 5 **13.** 3